

Context length

Context length or context window = is the maximum amount of text, measured in tokens--an AI model can process and remember at once. It acts as the model's short-term working memory, storing your prompts, conversation history, and any uploaded documents.

How it works

- **Tokens:** An AI model don't read words readily. They break text into "tokens" (roughly 3 to 4 characters or 3/4 of an English word)
- **The limit:** The context length represents the total limit for both your input and output
- **Forgetting:** Once a conversation or document exceeds the limit, older information is either permanently dropped or summarized to make room for new input.

Context capacities range from a few pages of text to entire novels or extensive codebases:

- **Mid-Size & Local Models:** 8,000 to 128,000 tokens (approx. 6 to 96 pages).
- **Advanced Models:** 200,000 to 1,000,000+ tokens (approx. 150 to 750+ pages).

Trade-off of larger contexts

While a massive context window allows you to upload entire books or complex code repositories at once, it comes with trade-offs:

- **Accuracy:** "Needle-in-a-haystack" testing shows that models can sometimes overlook specific facts buried in massive documents.
- **Performance:** Processing larger contexts requires more computational power, leading to slower response times and potentially higher costs.